Ex. No.:9 **Roll No.: 22BCS091**
Date :02.04.25

Component and Deployment Diagrams

Aim

To create Component and Deployment diagrams for Centralized Hostel Management system using CASE tools

Component Diagram

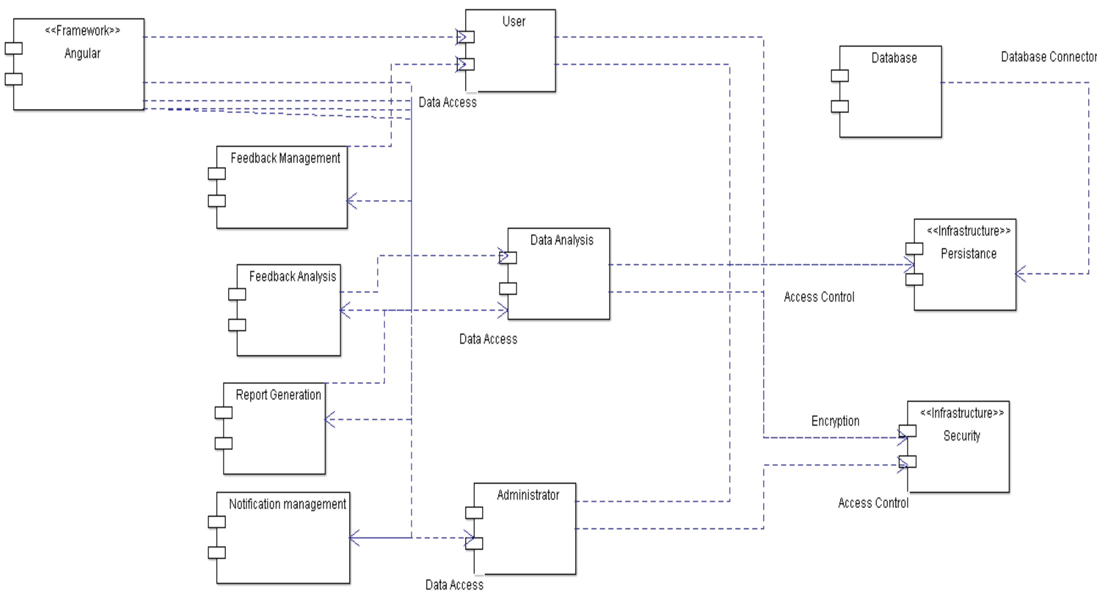

Fig: 9.1 – Component Diagram for Smart Feedback Collection and Analysis System

Description

Figure 9.1 shows the individual components of the **Feedback Management System**, depicted as a component diagram in the context of Object-Oriented Analysis and Design (OOAD). This diagram illustrates the architectural structure and interactions between various components of the system, emphasizing their distinct roles in achieving comprehensive functionality.

• **Component 1 - Angular Framework:**
This is the front-end framework used to build the user interface of the application. Angular communicates with the backend services to display and manage user interactions like feedback submission, notifications, and reports.

• **Component 2 - User:**
The User component represents the end-users who interact with the system through the Angular interface. They provide feedback and can view notifications and reports. It also interacts with backend services for data access and retrieval.

• Component 3 - Administrator:
This component manages backend operations like overseeing feedback data, generating reports, and sending notifications. It has broader data access privileges compared to regular users.

• **Component 4 - Feedback Management:**
Handles the collection and organization of user feedback. This component ensures that feedback is stored in a structured format, making it easy to access for analysis.

• **Component 5 - Feedback Analysis:**
Processes the feedback data collected by the system. It identifies patterns or issues based on user input, helping administrators make informed decisions.

• **Component 6 - Report Generation:**
Generates summaries and reports from the feedback analysis. These reports can be visual or textual and are used by administrators for decision-making and performance tracking.

• **Component 7 - Notification Management:**
Responsible for creating and distributing notifications to users based on certain triggers (e.g., feedback results, system updates). Ensures users are informed in real-time.

• **Component 8 - Data Analysis:**
This central component accesses and processes data for feedback analysis, report generation, and notifications. It works closely with the persistence and security layers for secure and efficient data handling.

• **Component 9 - Persistence (Infrastructure):**

Responsible for storing and retrieving data from the database. It ensures data durability and handles all interactions with the Database component.

• **Component 10 - Security (Infrastructure):**

Provides encryption and access control services to secure the application. It ensures that only authorized users and services can access sensitive data.

• **Component 11 - Database:**

A storage component that maintains all system data, including feedback, user information, reports, and notifications. It connects to the Persistence layer to support CRUD operations.

Deployment Diagram

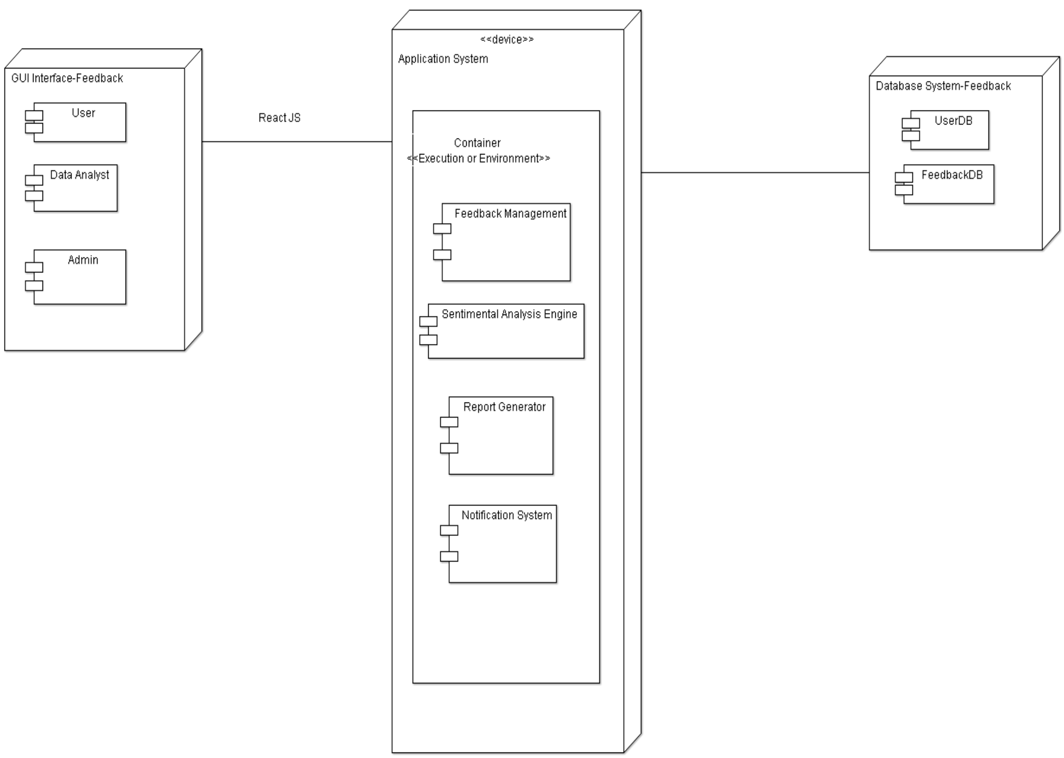

Fig: 9.2 – Deployment Diagram for Centralized Hostel Management system

Description

Figure 9.2 illustrates the component architecture of the **Feedback Analysis System**, modeled using a component diagram in the context of Object-Oriented Analysis and Design (OOAD). This diagram showcases the interaction among three primary systems: the Graphical User Interface (GUI), the Application System, and the Database System.

Device 1 – GUI Interface: Feedback

The first device, labeled GUI Interface: Feedback, serves as the front-end interface of the Feedback Analysis System. Developed using React JS, this interface enables user interaction with the system. It includes components for three types of users: the User, who submits feedback; the Data Analyst, who reviews trends and sentiment reports; and the Admin, who manages user access and system operations. This device captures user input and presents processed data, acting as the primary access point to the backend services.

Device 2 – Application System

The second device, referred to as the Application System, represents the backend environment where the system's main logic executes. This device contains a container marked as Execution or Environment, which hosts the following functional components:

- Feedback Management: Handles the submission, categorization, and storage of feedback.
- Sentimental Analysis Engine: Analyzes the sentiment of the feedback using natural language processing (NLP).
- Report Generator: Compiles the processed feedback into structured reports for administrative and analytical review.
- Notification System: Sends alerts and system notifications based on feedback trends or pre-defined rules.
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Device 3 – Database System: Feedback

The third device, named Database System: Feedback, is responsible for all data storage and retrieval operations. It includes two main components:

- UserDB: Stores user credentials, profiles, roles (User, Admin, Analyst), and session data.
- FeedbackDB: Stores all submitted feedback, sentiment analysis results, and related metadata such as timestamps and feedback categories.

This device ensures the persistence and integrity of data, and is tightly integrated with the Application System for real-time access and update.

Result

Thus the component and deployment diagrams for Smart Feedback Collection and Analysis is created using CASE tools.